ONTOLOGICAL THEORY OF
REALITY TOTALITIES
(TRT)

A Topological–Informational Framework for Quantum Gravity
Tribute to Carlo Rovelli

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Abstract

We present TRT, a framework for quantum gravity grounded in informational thermodynamics. The modular Hamiltonian $K_{\rho_n} = -\ln(\rho_n)$ of each node's reduced state is identified as the physical Hamiltonian at Planck scale (Postulate II), yielding the Energy–Information Identity $\langle H_n \rangle = S(\rho_n)$ in Planck units (Proposition 1). Quantum coherence is measured relative to a node-local geometric reference basis $\mathscr{B}_n$ — independent of the state — producing a non-trivial decomposition into classical information I_{cl} (associated with mass) and geometric coherence $C_{\mathscr{B}}$ (associated with the radiative sector). Spacetime emerges selectively on condensation channels (*zarcillos*) of a spin network. We prove one algebraic identity, introduce testable conjectures on cosmic topology, and rigorously separate algebraic results, physical hypotheses, and speculative interpretations.

Keywords: quantum gravity, modular Hamiltonian, thermal time, relative entropy of coherence, geometric basis, spin network condensation, fractal topology, CMB Cold Spot

1. Introduction

The persistent tension between general relativity and quantum theory stems from incompatible treatments of spacetime. Following Rovelli's relational programme [8] and the algebraic approach to QFT, we propose that both spacetime and time emerge from a more fundamental informational ontology on spin networks.

This paper does not claim a new mathematical theorem. The algebraic identity $\mathrm{Tr}(\rho(-\ln\rho)) = S(\rho)$ is standard; what we call the 'Energy–Information Identity' is the statement that, once $H_n := k_B T_P K_{\rho_n}$ is taken as the physical Hamiltonian, its expectation value equals von Neumann entropy in Planck units. The originality lies in the ontological and thermodynamic consequences.

We separate three levels: (a) **algebraic identities** from definitions, (b) **physical hypotheses** requiring Postulate II, and (c) **conjectures** that are physically motivated but quantitatively open.

2. Postulates

2.1 Postulate I — Adimensional Ontology

Reality consists of adimensional points organized in a spin-network graph $\Gamma = (V, E)$ with finite-dimensional local Hilbert spaces $\mathscr{H}_n$. Spacetime is emergent.

State Hierarchy

Global: $|\Psi\rangle \in \mathscr{H} = \otimes_n \mathscr{H}_n$ is a pure state.

Local: $\rho_n = \mathrm{Tr}_{V\setminus\{n\}}(|\Psi\rangle\langle\Psi|)$ is the reduced density matrix (generically mixed).

Faithfulness: We **assume** each ρ_n is faithful (full-rank) on $\mathscr{A}_n$. This is a technical assumption: reduced states can have zero eigenvalues if Schmidt rank < $\dim(\mathscr{H}_n)$. Faithfulness ensures $-\ln(\rho_n)$ is well-defined. For non-faithful states, regularize via $\rho_n^{\varepsilon} = (1-\varepsilon)\rho_n + \varepsilon I/d_n$, $\varepsilon \to 0$.

2.2 Postulate II — Modular Hamiltonian as Physical Hamiltonian

Postulate II (Physical Identification)

For each node n with faithful ρ_n, the local physical Hamiltonian is:

$$H_n = k_B T_P \cdot K_{\rho_n} \qquad where \;\; K_{\rho_n} = -ln(\rho_n)$$

Physical time at node n is the local modular flow rescaled by t_P. T_P is the fundamental calibration point, not the current cosmological temperature.

Justification for T_P: (a) Universality — unique temperature from ħ, c, G, k_B. (b) Landauer — erasing one nat costs E_P. (c) KMS — ρ_n satisfies KMS at $\beta = 1$ w.r.t. modular time; rescaling yields $\beta_{phys} = 1/(k_B T_P)$.

2.3 Postulate III — Informational Redistribution

Postulate III (Reformulated — v3.4)

Under unitary evolution of $|\Psi\rangle$, the **aggregate local modular energy** $E_{agg} = \Sigma_n S(\rho_n)$ is an emergent functional that **changes** as global dynamics redistributes entanglement among nodes. E_{agg} is not a conserved quantity: a global unitary can create, redistribute, or reduce correlations between specific subsystems.

What unitarity **does** preserve is the global purity ($S(|\Psi\rangle\langle\Psi|) = 0$) and the global spectrum of ρ_{global}. The physical content of Postulate III is **phenomenological**: in the regime described by TRT, the dominant processes (diffusion, crystallization, BH–WH cycles) *redistribute* E_{agg} among nodes without net long-term creation or destruction, because these processes are approximately entropy-preserving at the aggregate level. This is a dynamical hypothesis about the effective physics, not a consequence of unitarity alone.

2.4 Postulate IV — Dissipation Principle

The universe is the thermodynamic solution to dissipating maximal non-crystallizable information without collapse, via: (a) cosmic expansion, (b) Higgs-mediated materialization, (c) fractal recursive BH–WH cycles.

3. Mathematical Setup

3.1 Tomita–Takesaki Modular Theory [2, 11]

Given $(\mathscr{A}, \mathscr{H}, \Omega)$ where Ω is cyclic and separating, the Tomita operator S has polar decomposition $S = J\Delta^{1/2}$. The modular flow $\sigma_t^\rho(x) = \Delta^{it}x\Delta^{-it}$ is a *-automorphism of $\mathscr{A}$, and ρ satisfies KMS at $\beta = 1$ [4].

3.2 Scope: Finite Spin Networks

Remark (Type I restriction)

We work on finite spin networks with $\dim(\mathscr{H}_n) < \infty$, giving Type I algebras where $S(\rho_n)$ is finite [12, 15].

3.3 Von Neumann Entropy

$$S(\rho) = -Tr(\rho \ln \rho) \qquad [nats] \quad (3.1)$$

3.4 Coherence Relative to a Geometric Reference Basis

Definition 1 (Geometric Basis and Coherence) — v3.4

Each node n of the spin network carries a **node-local geometric reference basis** $\mathscr{B}_n$, defined as a recoupling/intertwiner basis compatible with the local geometric observables (area, volume) of LQG [3]. In the simplest models, $\mathscr{B}_n$ reduces to a spin basis $\{|j,m\rangle\}$; in the full theory, it is the intertwiner basis of the node's gauge-invariant Hilbert space. The key property is that $\mathscr{B}_n$ is **state-independent**: it does not depend on ρ_n.

The dephasing channel in $\mathscr{B}_n$ is $\Delta_{\mathscr{B}}(\rho_n) = \Sigma_k \langle k|\rho_n|k\rangle\, |k\rangle\langle k|$ where $\{|k\rangle\} \in \mathscr{B}_n$.

The **geometric coherence** is:

$$\mathbf{C_{\mathscr{B}}(\rho_n) = S(\Delta_{\mathscr{B}}(\rho_n)) - S(\rho_n) \geq 0}$$

Non-negative because $\Delta_{\mathscr{B}}$ is a unital channel (Schur–Horn majorization).

Remark (Separation of roles — v3.3–3.4)

K_{ρ_n} defines **energy** (Proposition 1). $\mathscr{B}_n$ defines **coherence**. These are different bases whenever ρ_n has off-diagonal elements in $\mathscr{B}_n$, which is generic for entangled states. $C_{\mathscr{B}}(\rho_n)$ measures how much quantum information is misaligned with the geometric degrees of freedom.

Remark (Low-temperature limit — per DeepSeek R4)

At very low temperatures, $K_{\rho_n} \approx \beta H_n$. If H_n is approximately diagonal in $\mathscr{B}_n$ (a common assumption in LQG where the Hamiltonian is a function of area/volume operators), then the two bases approximately coincide and $C_{\mathscr{B}}(\rho_n) \to 0$ for thermal states. This is physically correct: at low T, thermal states are nearly diagonal in the geometric basis. The interesting regime for $C_{\mathscr{B}}$ is intermediate energies: BH evaporation, the early universe, or any process where quantum superpositions of distinct geometries are significant.

Classical information in the geometric basis:

$$I_{cl}(n) := S(\Delta_{\mathscr{B}}(\rho_n)) \qquad I_{cl}(n) = S(\rho_n) + C_{\mathscr{B}}(\rho_n) \quad (3.2–3.3)$$

Remark (I_cl includes thermal entropy — per DeepSeek R4)

$I_{cl}(n)$ measures the total entropy of the geometry-dephased state, which includes both classical information associated with mass and any thermal entropy of crystallized degrees of freedom. The identification with mass is strongest for the zero-temperature ground state manifold; at finite temperature, I_{cl} also includes thermal excitations.

4. The Energy–Information Identity

Proposition 1 (Energy–Information Identity)

For faithful ρ_n with $H_n := k_B T_P K_{\rho_n}$, in Planck units:

$$\langle H_n \rangle = S(\rho_n)$$

Proof. $H_n = -\ln(\rho_n)$ in Planck units. $\langle H_n \rangle = \mathrm{Tr}(\rho_n(-\ln\rho_n)) = S(\rho_n)$.

■

Corollary. The 44% discrepancy in v1.0 = $1/\ln(2) \approx 1.4427$ (nats/bits).

Remark (Aggregate energy)

$E_{agg} = \Sigma_n S(\rho_n)$ is the sum of local entanglement entropies, **not** the global entropy (zero for pure $|\Psi\rangle$). By subadditivity, $E_{agg} \geq 0$ with equality only for product states. E_{agg} is not conserved under arbitrary global unitaries; its variations are constrained by global purity and subadditivity inequalities. Whether E_{agg} contributes directly to emergent curvature or overcounts via mutual information is connected to P1.

4.1 Heuristic Correspondences

Dimensional and calibration checks (not independent derivations):

Bekenstein–Hawking [6, 7]. $S_{BH} = A/(4l_P^2)$. Under TRT: $E_{BH} = k_B T_P \cdot S_{BH}$. Dimensionally consistent with Mc^2, provided S_{BH} equals aggregate modular information of horizon nodes. Heuristic.

Landauer [5]. Erasing one nat costs E_P at $T = T_P$.

Holographic. $I \leq A/(4l_P^2) \rightarrow E \leq A/(4l_P^2)$ in Planck units.

Unruh. $T_U = T_P \cdot al_P/(2\pi c^2)$. Dimensionally correct.

5. Informational Thermodynamics

5.1 The Mass–Radiation Dictionary

Interpretation (phenomenological mapping)

I_{cl} ↔ Mass (crystallized information). The diagonal part of ρ_n in $\mathscr{B}_n$ represents information aligned with geometric degrees of freedom (area, volume eigenvalues). Massive particles correspond to nodes where ρ_n is approximately diagonal in $\mathscr{B}_n$.

$C_{\mathscr{B}}$ ↔ Radiative sector (non-crystallizable information). Off-diagonal elements of ρ_n in $\mathscr{B}_n$ represent information misaligned with geometry — quantum superpositions of distinct geometric configurations. The radiative sector is **dominantly coherent**, not exclusively so: realistic radiation states may retain a small diagonal component. The identification is phenomenological.

5.2 The BH–WH Cycle

BH absorption: Matter (I_{cl}) falls in; Planck-scale dynamics dissolves diagonal structure in $\mathscr{B}$ → $C_{\mathscr{B}}$ increases.

BH emission: Hawking radiation emits predominantly coherent quanta. Page curve describes gradual transfer.

Recrystallization: Coherence recondenses into diagonal form in $\mathscr{B}$ (new stable geometric configurations).

Remark (Redistribution, not conservation)

Postulate III asserts that the dominant physical processes redistribute E_{agg} among nodes without net long-term creation or destruction. This is a dynamical hypothesis, not a consequence of unitarity. The local redistribution between I_{cl} and $C_{\mathscr{B}}$ at each node is governed by Eq. 7.1.

6. Fractal Topology: Zarcillos and Océanos

6.1 The Condensation Function

Definition 2 (Condensation Function)

$c: V \to [0, 1]$, $c(n) = 1 - \exp(-I_{cl}(n)/I_P)$ where $I_P = 1$ nat (Planck information unit).

Remark (Connection to LQG spin labels — per DeepSeek R4)

A natural ansatz is $I_{cl}(n) \propto \Sigma_{e \ni n} j_e(j_e+1)$, where the sum runs over edges incident to n and j_e are the spin labels. This makes c(n) a function of local geometric data, consistent with crystallized information being encoded in definite spin configurations. This ansatz is speculative and awaits verification (P3).

6.2 Emergent Topology

Definition 3 (Zarcillos and Océanos)

Zarcillos: $\Gamma_z = \{n : c(n) > c_{crit}\}$. Metric $g_{\mu\nu}$ emerges via LQG coarse-graining.

Océanos: $\Gamma_o = \{n : c(n) \leq c_{crit}\}$. Metric undefined (absence of geometric description).

6.3 Physical Correspondence

Scale	Zarcillo	Océano
Planck	High-spin nodes	Low-spin / disconnected
Nuclear	Decay channels	Tunneling barriers
Stellar	Accretion, jets	Interstellar vacuum
Galactic	Spiral arms, bars	Halos
Cosmic	Filaments (cosmic web)	Voids (Eridanus)

Conjecture 1 (Fractal Condensation) [qualitative]

c(n) exhibits statistical self-similarity; zarcillo sizes follow $P(\text{size} > s) \sim s^{-\alpha}$, $\alpha \in [1.5, 2.5]$. Requires simulation (P3).

7. Information Diffusion and Cosmic Expansion

7.1 Diffusion on the Graph

Let $x(t) \in \mathbb{R}^{|V|}$ with $x_n(t) = C_{\mathscr{B}}(\rho_n(t))$.

$$dx_n/dt = -D\,(L_\Gamma x)_n - \kappa_n\, x_n(1 - x_n/x_{max}) + s_n \quad (7.1)$$

Remark (Saturation and well-posedness — v3.4)

In finite-dimensional $\mathscr{H}_n$, $C_{\mathscr{B}}(\rho_n)$ is bounded above by $x_{max} = \ln(d_n)$ nats, where $d_n = \dim(\mathscr{H}_n)$. The logistic factor $(1 - x_n/x_{max})$ in the crystallization term ensures the ODE respects this physical bound: crystallization rate vanishes as coherence approaches its maximum. Without it, the linear ODE could push x_n beyond the physical range.

Eq. 7.1 is a mesoscopic approximation: $C_{\mathscr{B}}(\rho_n)$ is coarse-grained to a scalar, effectively assuming that the phase information of off-diagonal elements is randomized on short timescales (standard in open quantum systems → rate equations). Deriving Eq. 7.1 from spinfoam dynamics is P5.

7.2 Dissipation Principle

Proposition 2 (Information Dissipation)

Modular flow is a *-automorphism (no entropy production). Effective coarse-grained dynamics can be modeled as CPTP channels Φ_t, for which:

$$D(\Phi_t(\rho) \,//\, \Phi_t(\sigma)) \leq D(\rho \,//\, \sigma)$$

Monotonic loss of distinguishability under coarse-graining.

Remark (Local time synchronization)

Each node has its own modular clock. On zarcillos, local flows synchronize via coarse-graining, recovering a global time coordinate. On océanos, time remains purely local and graph-theoretic.

7.3 Expansion as Hilbert Space Growth

Conjecture 2 (Thermodynamic Expansion) [quantitatively open]

When total $C_{\mathscr{B}}$ exceeds $C_{max} \sim 1$ nat per node, node creation occurs. In the continuum limit, $N \sim a(t)^3$. If $\langle C_{\mathscr{B}} \rangle \propto \rho_{coh} \cdot a^3$, then $d(\ln a)/dt \sim \gamma(\rho_{coh} - \rho_{crit})$, mimicking Friedmann with effective Λ when $\rho_{coh} \to$ const. Recovering $\Lambda \sim 10^{-122}$ requires $\rho_{coh}(\text{today}) \sim 10^{-122}$ in Planck units. Rate $\gamma \sim \omega_P$. This is P1.

8. The Crystallization Epoch and the CMB

8.1 Cosmic Timeline

Epoch	Standard	TRT interpretation	I_cl / C_D
$t < 10^{-36}$ s	Planck era	Full modular regime	0 / max
$\sim 10^{-12}$ s	QGP → hadrons	QCD crystallization	↑ / ↓
~3 min	BBN	Nuclear crystallization	~0.25 / ~0.75
~380 kyr	Recombination	Massive crystallization	~0.31 / ~0.69
Today	Structure	Crystallization + BH decrystallization	↑ / ↓

8.2 CMB as Primordial Coherence

Interpretation (phenomenological restatement)

CMB photons are primordial $C_{\mathscr{B}}$ decoupled at recombination. Anisotropies encode c(n) statistics. Restatement, not prediction.

8.3 The Cold Spot

Conjecture 3 (Cold Spot) [falsifiable]

The Cold Spot (Eridanus, ~5°, $\Delta T \sim -70$ µK) is a primordial océano. Standard ISW-supervoid explains ~10–20%; TRT offers complementary topological explanation.

Falsifiable: lower topological connectivity (fewer filaments, lower Betti numbers) vs. density-matched controls. Testable via persistent homology of DESI/Euclid galaxy distributions [16]. If connectivity is normal, this conjecture falls.

9. Emergence of $E = mc^2$

Conjecture 4 (Mass from Modular Eigenvalues) [interpretive]

At $T \ll T_P$, decoherence selects $\mathscr{B}_n$ and K_{ρ_n} develops stable eigenvalues $\lambda_k \to m_k = k_B T_P \lambda_k / c^2$. Higgs couplings determine stability. Physical conjecture (P2).

Electron: $\lambda_e \approx 4.18 \times 10^{-23}$ nats. Mass = frozen I_{cl} — information aligned with geometry.

10. Classification of Results

Result	Type	Status	Sec.
⟨H_n⟩ = S(ρ_n)	Proposition	Algebraic identity	4
ln(2) factor	Corollary	Proven	4
BH / Landauer / Unruh	Heuristic	Dimensional checks	4.1
Geometric coherence C_D(ρ_n)	Definition	Rigorous (standard QI)	3.4
Mass ↔ I_cl, Radiation ↔ C_D	Interpretation	Phenomenological mapping	5.1
Zarcillos / Océanos	Definition	Rigorous	6
I_cl ∝ Σ j(j+1) ansatz	Conjecture	Speculative ansatz	6.1
Fractal power-law	Conjecture	Requires simulation (P3)	6.4
Diffusion Eq. 7.1 (saturated)	Proposition	Well-posed ODE; params TBD	7.1
Dissipation (data processing)	Proposition	Standard QI	7.2
Expansion from coherence	Conjecture	Quantitatively open (P1)	7.3
CMB = primordial C_D	Interpretation	Restatement	8.2
Cold Spot = min condensation	Conjecture	Falsifiable (P4)	8.3
Mass from modular eigenvalues	Conjecture	Deepest open (P2)	9

10.1 Open Problems

P1 (Critical). Derive $\Lambda \sim 10^{-122}$ from Conjecture 2.

P2 (Critical). Map modular spectrum to SM masses.

P3 (Feasible). Simulate Eq. 7.1 on random graphs; test Conjecture 1 and I_{cl} ansatz.

P4 (Feasible). Test Conjecture 3 via DESI/Euclid persistent homology.

P5 (Foundational). Derive diffusion parameters from spinfoam → Lindblad.

P6 (Foundational). Formalize Higgs on spin network.

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Appendix A: Changelog v3.3 → v3.4

Structural fixes (ChatGPT R4):

(1) Postulate III reformulated: E_{agg} is an emergent functional that **changes** under global dynamics, not conserved. Unitarity preserves global purity/spectrum, not E_{agg}. Redistribution is a dynamical hypothesis. (2) Geometric reference basis generalized from $\{|j,m\rangle\}$ to abstract node-local intertwiner basis $\mathscr{B}_n$, avoiding toy-model LQG objection. (3) Radiative sector: 'dominantly coherent' replaces 'purely off-diagonal'. (4) ODE 7.1 gains logistic saturation factor $(1 - x_n/x_{max})$ ensuring physical bound $x_n \leq \ln(d_n)$.

Additions (DeepSeek R4):

(5) Low-temperature limit remark: bases converge, $C_{\mathscr{B}} \to 0$ at low T (physically correct). (6) I_{cl} includes thermal entropy remark. (7) Ansatz $I_{cl}(n) \propto \Sigma\, j_e(j_e+1)$ connecting condensation to LQG spin labels. (8) Phase randomization remark for scalar ODE approximation.

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In tribute to Carlo Rovelli.